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Governance in Decentralized Finance: Aligning DAOs with IT Management and Regulatory Compliance in Indonesia

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***Abstract*—This paper explores the alignment between DAO governance mechanisms, IT governance frameworks, and Indonesia's regulatory environment. Using a qualitative document analysis of academic literature, DAO governance documentation, IT governance standards (COBIT, ITIL, ISO/IEC 38500), and Indonesian regulatory texts issued by Bappebti, Bank Indonesia, and the Financial Services Authority (OJK), the study identifies critical areas of convergence and divergence. Findings indicate that DAOs demonstrate strong transparency due to on-chain records, but remain weak in accountability, stakeholder alignment, risk management, and compliance. These weaknesses generate friction with Indonesia's fragmented regulatory landscape, where crypto assets are classified as commodities, prohibited as payment instruments, and inconsistently taxed. Stakeholder management emerges as a pressing concern, as token-weighted voting privileges large holders while Indonesian retail investors face governance literacy gaps. To address these challenges, the paper proposes a hybrid governance framework that integrates IT governance principles with DAO structures, supplemented by compliance mechanisms tailored to Indonesian regulations. Recommendations include the introduction of accountability representatives, adoption of quadratic voting, adaptation of COBIT-based risk management, and establishment of regulatory sandboxes. The study contributes to the discourse on DeFi governance by offering a conceptual model for reconciling decentralized innovation with national regulatory frameworks in emerging markets such as Indonesia.**

Keywords—decentralized finance, decentralized autonomous organizations, IT governance, cobit, indonesia

I. Introduction

The rise of Decentralized Finance (DeFi) and Decentralized Autonomous Organizations (DAOs) has significantly transformed the global financial sector by using blockchain and smart contracts to provide financial services outside of traditional intermediaries, with DeFi protocols exceeding USD 40 billion in total value locked by 2023 [1, 2, 3]. However, this rapid growth has brought forth critical challenges related to governance, accountability, and regulatory compliance, issues central to effective information technology (IT) management [4]. DAOs, which manage financial protocols through smart contracts and collective decision-making, face their own governance challenges, including concentrated voting power, limited participant literacy, and unclear accountability [5]-[8]. These issues echo concerns in IT governance, such as accountability and risk control, yet demand specialized solutions for decentralized environments [9].

Indonesia, with its rapidly growing digital economy projected to reach USD 146 billion by 2025 and over 17 million crypto investors in 2023, is a global leader in cryptocurrency adoption, making the relevance of DeFi and DAOs particularly pronounced [10]-[12]. DeFi offers new investment avenues and promotes financial inclusion for the 40% of unbanked adults in rural and semi-urban areas [13]. Despite this enthusiasm, regulatory clarity has lagged, placing DAOs and DeFi in a governance "gray zone." Indonesia's fragmented regulatory framework reflects this tension: crypto assets are overseen by Bappebti, while OJK focuses on fintech but not directly on DeFi, and Bank Indonesia prohibits crypto as a payment instrument even as it develops the Digital Rupiah [14]-[16]. This highlights a paradox between Indonesia's innovation goals and its traditional centralized oversight.

From an IT governance standpoint, this regulatory ambiguity is crucial, as established frameworks like COBIT, ITIL, and ISO/IEC 38500 emphasize structured accountability, strategic alignment, and compliance, which DAOs typically lack due to their distributed authority across token holders and jurisdictions [17]. This raises a key research question: how can IT governance principles be adapted to Indonesian DAO ecosystems to enhance accountability, stakeholder alignment, and compliance without compromising decentralization [18][19]? The diverse stakeholders in Indonesian DeFi, including retail investors, exchanges, fintech startups, regulators, and developers, often have conflicting priorities, exacerbated by low financial literacy among retail investors, which increases risks of governance apathy and manipulation [20]. Furthermore, the lack of integrated anti-money laundering (AML) and know-your-customer (KYC) mechanisms in DAOs creates compliance gaps within Indonesia's financial integrity framework [21]. These challenges underscore the urgent need for a hybrid governance framework that integrates DAO decentralization with IT governance principles and Indonesia's specific regulatory requirements.

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II. LITERATURE RIVIEW

A. Decentralized Finance (DeFi): Global and Indonesian Perspectives

Decentralized Finance (DeFi) represents a blockchain-based financial ecosystem that facilitates transactions without traditional intermediaries, relying instead on self-executing smart contracts, primarily on Ethereum. Since 2020, DeFi has seen substantial global growth, with its Total Value Locked (TVL) reaching over USD 100 billion in late 2021 before stabilizing at around USD 40 billion in 2023 [3]. This expansion has led to diversification into various mechanisms such as decentralized exchanges (DEXs), stablecoins, and yield-generating activities like liquidity mining and staking [7][23]. While offering economic opportunities, DeFi also presents regulatory challenges and risks, including market instability and governance vulnerabilities within decentralized autonomous organizations (DAOs) [4][8].

In Indonesia, DeFi adoption mirrors the broader interest in cryptocurrencies, with over 17 million registered crypto users by 2023, making it one of the largest retail investor bases globally [11][24]. Domestic exchanges like Indodax and Tokocrypto have supported this growth, though participation has largely been speculative rather than focused on DeFi governance [20]. Despite this, interest in decentralized platforms is growing among younger, tech-savvy populations in major cities, leading to increased academic and policy discussions about DeFi's transformative potential [17][18]. A significant opportunity for DeFi in Indonesia lies in its ability to enhance financial inclusion, offering low-barrier access to credit, savings, and investments for the roughly 40% of the adult population that remains unbanked or underbanked, aligning with the country's digital economy ambitions [13][25]-[27].

However, the promise of DeFi in Indonesia is tempered by persistent challenges. Low financial literacy among retail users exposes them to risks such as scams and asset mismanagement, exacerbated by the inherent volatility of DeFi markets [28][29]. Furthermore, there is a regulatory disconnect: DeFi's global and decentralized governance through DAOs contrasts with Indonesia's territorial and fragmented regulatory landscape. While crypto assets are classified as commodities under Bappebti, Bank Indonesia prohibits their use as payment instruments, and the Financial Services Authority (OJK) primarily oversees fintech rather than specific DeFi regulations [14][15]. New taxation rules introduced in 2022 further complicate compliance for global DeFi protocols, creating uncertainty for both local users and developers [16]. These dynamics highlight the need to examine DeFi not just as a financial innovation but as a critical IT governance issue in Indonesia, suggesting that bridging DAO governance with IT governance frameworks like COBIT or ISO/IEC 38500 could help balance decentralization with accountability and compliance [9][30].

B. Decentralized Autonomous Organizations (DAOs)

Decentralized Autonomous Organizations (DAOs) represent a novel organizational paradigm within the broader DeFi ecosystem, operating as blockchain-based entities governed by smart contracts and collective decision-making to minimize reliance on centralized authorities while ensuring transparency and immutability in governance [4][31]. Members hold governance tokens that confer voting rights, allowing them to propose and approve protocol changes, allocate treasury resources, or adjust governance parameters [5]. Prominent examples like MakerDAO and Uniswap DAO demonstrate how decentralized governance can coordinate global communities without traditional corporate hierarchies [6][22]. However, DAOs face several governance risks, including the potential for plutocracy due to token-weighted voting, voter apathy from low incentives or mistrust, and vulnerabilities to coding flaws as exemplified by the 2016 "DAO hack" [7][8][18][23][32]. Furthermore, the lack of consistent legal recognition across jurisdictions complicates accountability, liability, and compliance, leaving DAOs in a regulatory gray zone [21][30].

In Indonesia, DAOs are an emerging and underexplored phenomenon, with local experiments primarily associated with NFT communities, gaming guilds, and creative economy collectives operating informally and borrowing from cooperative or association models [20]. This is due to Indonesian law currently recognizing entities like Perseroan Terbatas (limited liability companies) or koperasi (cooperatives), but not decentralized organizations [14]. This legal gap raises fundamental questions about the future recognition and alignment of DAOs with Indonesia's governance and regulatory frameworks. Without legal clarity, DAOs in Indonesia face constraints in legitimacy, enforceability, and integration into formal financial systems, despite attracting growing interest from blockchain innovators and creative sectors.

C. IT Governance Models

IT governance, through frameworks like COBIT, ITIL, and ISO/IEC 38500, provides structured guidance for managing IT resources, aligning them with organizational objectives, and ensuring compliance. COBIT focuses on accountability, transparency, and risk management by aligning IT with business strategy [17], while ITIL emphasizes best practices for IT service management, reliability, and continuous improvement [9]. ISO/IEC 38500 offers corporate governance standards for IT, covering responsibility, strategy, acquisition, performance, conformance, and human behavior [33]. These frameworks are widely implemented in Indonesia, particularly in government and financial sectors, driven by regulations from the Financial Services Authority (OJK) and Bank Indonesia [14], which foster structured IT governance for accountability and risk control in conventional organizations [11].

However, the emergence of Decentralized Autonomous Organizations (DAOs) challenges these traditional, hierarchical governance models due to their distributed accountability and lack of executive structures [4]. Existing frameworks, designed for clear authority and responsibility, do not adequately address DAO-specific risks such as smart contract vulnerabilities, governance capture, and voter apathy [7][23]. Therefore, adapting IT governance principles for DAOs is crucial. This involves reinterpreting COBIT's focus to align DAO governance with token holders' collective interests and regulatory compliance [20], and reframing ISO/IEC 38500's emphasis on accountability and performance to address transparency and trust in decentralized decision-making, especially in contexts like Indonesia where centralized oversight remains the regulatory norm [15].

D. Stakeholder Management in DAOs

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Stakeholder management is critical for organizational sustainability. In DAOs, stakeholders include developers, token holders, liquidity providers, users, and increasingly, regulators. However, DAOs face unique stakeholder challenges:

TABLE I. STAKEHOLDER'S CHALLENGES IN DAOS

Challenge	Description	Sources
Power asymmetry	Token distribution often favors early investors or venture capital firms, leading to centralization within decentralized governance.	[4] [7] [32]
Lack of participation`	Retail users often do not engage in governance due to lack of incentives or technical barriers.	[18] [20] [23]
Cross-jurisdictional complexity	DAO stakeholders may be globally dispersed, complicating consensus and regulatory obligations.	[4] [8] [15] [21]

Power asymmetry significantly impacts crypto adoption in Indonesia, where retail participants primarily engage with speculative motives rather than governance [11]. This makes the Indonesian market vulnerable to global "whales" or venture capital firms that hold large governance token holdings in international DAOs, potentially leading to decisions that do not represent Indonesian stakeholders and undermining decentralization's inclusive ideals. As in [7] note, token distribution often replicates centralized power structures, increasing the risk of governance capture. This dynamic challenges DAOs' potential as community-driven governance mechanisms, especially in Indonesia, where the state prioritizes digital sovereignty and local stakeholder protection. Furthermore, a lack of participation in Indonesia stems from low financial and governance literacy among retail investors. Despite Indonesia having over 17 million crypto users, one of the highest global adoption rates [24], most engage in short-term trading instead of long-term protocol governance [20]. Studies indicate that retail users often lack sufficient incentives or technical capacity for complex DAO governance [23]. This apathy is reinforced by cultural and structural barriers, with Indonesian investors often trusting centralized exchanges like Indodax or Tokocrypto as intermediaries rather than engaging directly in decentralized governance. [28] highlights that this creates vulnerability to scams and misinformation, increasing governance manipulation risks. Without mechanisms to enhance literacy and incentivize active participation, DAOs in Indonesia may fail to achieve democratic decision-making and accountability.

Cross-jurisdictional complexity highlights the legal and regulatory uncertainties facing DAOs in Indonesia. While Bappebti regulates crypto assets as commodities, Bank Indonesia prohibits their use as payment instruments, and the Financial Services Authority (OJK) supervises fintech services, none directly recognize DAOs as legal entities [14]. This fragmented oversight creates compliance gaps, particularly concerning anti-money laundering (AML), know-your-customer (KYC), and taxation obligations. While [21] emphasize that global regulatory frameworks for decentralized finance remain unsettled, resulting in DAOs operating in a legal vacuum in Indonesia, [15] observe that central bank digital currency (CBDC) initiatives underscore the state's preference for centralized financial innovation models, contrasting with DAOs' borderless, decentralized nature. Consequently, Indonesian DAO participants face uncertainties regarding liability, taxation introduced in 2022, and alignment with national financial integrity standards. This regulatory paradox illustrates the urgent need for hybrid governance models that reconcile DAO practices with Indonesia's compliance and oversight structures.

E. Indonesia Regulatory Frameworks and IT Compliance

Indonesia's crypto landscape is complex, with Bappebti recognizing crypto assets as commodities and regulating centralized exchanges while largely overlooking DeFi and DAOs. Bank Indonesia prohibits cryptocurrency for payments but is developing its Digital Rupiah (Project Garuda) [35][36]. The OJK also lacks regulations for DeFi and DAOs, creating jurisdictional gaps. While VAT and income tax on crypto transactions since May 2022 signify government recognition, compliance is difficult for DAOs due to their lack of AML/KYC, conflicting with Indonesia's financial integrity obligations and exposing participants to regulatory risks [4][21]. The global nature of DAOs further complicates enforcement for national regulators, highlighting cross-jurisdictional governance complexities [7][8][37]. Existing literature highlights DeFi and DAOs' potential to revolutionize financial governance through smart contracts and community participation, promoting transparency and autonomy [4][5][22]. However, these models face challenges in accountability, risk management, and compliance, including governance capture by "whales" and voter apathy [18] [38]. Smart contract vulnerabilities and the absence of legal recognition expose DAO participants to risks without clear liability structures [39][40].

While IT governance frameworks like COBIT, ITIL, and ISO/IEC 38500 offer principles of accountability, transparency, and risk control [17][33] and are widely adopted in hierarchical organizations in Indonesia [14][34], their application to DAOs remains largely unexplored due to the decentralized nature of DAOs. The literature acknowledges the need to adapt IT governance principles to DAOs, but practical frameworks are scarce. Global studies have explored DAO governance challenges, but there is limited scholarship on how IT governance frameworks can be adapted for DAO ecosystems in emerging markets like Indonesia, especially in reconciling decentralized governance with fragmented national regulatory systems and unique socio-economic realities such as low financial literacy and the dominance of retail investors. This study aims to bridge this gap by proposing a hybrid DAO–IT governance model tailored for Indonesia, integrating IT governance principles—including accountability, transparency, risk management, and strategic alignment—into DAO operations while remaining responsive to Indonesia's regulatory environment and stakeholder needs.

III. RESEARCH DESIGN AND METHODOLOGY

This study employs a conceptual and exploratory qualitative document analysis [41][42] to investigate the intersection of Decentralized Finance (DeFi), Decentralized Autonomous Organizations (DAOs), and IT governance within the Indonesian context. The research aims to analyze the adaptation of IT governance principles to DAO ecosystems, explore stakeholder management challenges in DAO and regulatory documents, and assess how Indonesian regulatory frameworks interact with DAO governance. To achieve these objectives, the study is guided by three research questions focusing on the alignment/divergence of DAO and

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IT governance, insights into stakeholder management from relevant documentation, and how Indonesian regulations address DAO governance and compliance.

The study relies on documentary data from four categories: academic literature on DeFi, DAO structures, and IT governance; established governance frameworks such as COBIT 2019, ITIL 4, and ISO/IEC 38500; official DAO documentation from leading DAOs like MakerDAO and Uniswap; and key Indonesian regulatory and policy documents, including Bappebti, Bank Indonesia, OJK, and taxation regulations related to crypto assets. Data collection involves systematic literature searches and retrieval of official documents from reputable sources [43]. Qualitative Content Analysis (QCA) is then applied [44][45], involving coding documents into categories, clustering codes into thematic groupings, and interpreting themes to identify areas of convergence and divergence between DAO practices, IT governance principles, and Indonesian regulatory requirements.

To ensure validity and reliability, the research utilizes source triangulation, thematic consistency checks, and transparent documentation of processes. While acknowledging limitations such as the lack of empirical data on Indonesian DAO participants, the idealized nature of some DAO documents, the rapidly evolving regulatory landscape, and the tension between global DAO scope and territorial Indonesian regulations, the methodology is deemed suitable for the study's objectives [44]. By systematically analyzing documents across these diverse domains, the research aims to provide a robust foundation for proposing a hybrid governance model that integrates DAO principles with IT governance and regulatory compliance in Indonesia.

III. Discussion

A. DAO Governance vs. IT Governance Principles

Decentralized Autonomous Organizations (DAOs) offer a fundamentally new model of governance built on blockchain technology, but when compared with established IT governance frameworks such as COBIT, ITIL, and ISO/IEC 38500, their structural incompleteness becomes evident. While DAOs excel in transparency, they exhibit significant weaknesses in accountability, stakeholder alignment, risk management, and compliance. These gaps present challenges, particularly in jurisdictions like Indonesia, where regulatory expectations demand clearly defined responsibility and oversight at Fig. 1.

Decentralized Autonomous Organizations (DAOs) face significant challenges in accountability and stakeholder alignment when compared to established IT governance frameworks. While DAOs distribute decision-making, they lack clear accountability structures due to the absence of executive boards, creating ambiguity in assigning liability for failures such as governance manipulation or protocol hacks [4][5]. This contrasts sharply with IT governance frameworks like COBIT and ISO/IEC 38500, which mandate explicit accountability and board-level responsibility [17]. Furthermore, despite aiming for inclusivity, DAOs often suffer from power asymmetry, where token-weighted voting favors large holders, marginalizing smaller participants, especially in markets with limited governance literacy [7][11][20][32]. This gap in fair representation leads to a misalignment between community values and governance outcomes, differing from IT governance frameworks that prioritize aligning decisions with stakeholder value [17].

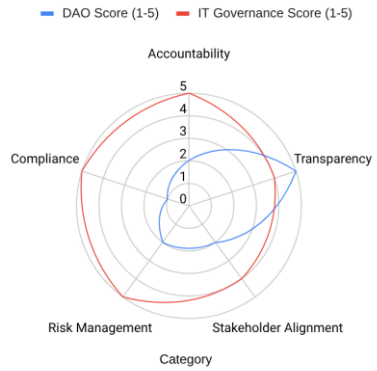

Fig. 1. DAO vs IT governance principles (comparative strengths).

Transparency is a key strength of DAOs, as all governance proposals, voting outcomes, and treasury transactions are immutably recorded on-chain and publicly accessible [22][23]. This inherent transparency often surpasses many centralized systems, ensuring open access to governance data. IT governance frameworks such as COBIT and ITIL also promote transparency for auditability and stakeholder trust [33]. This area demonstrates strong convergence between DAO and IT governance principles, presenting an opportunity for cross-learning and strengthening best practices in both domains.

However, DAOs exhibit considerable gaps in risk management and compliance. Risk management in DAOs is primarily reactive and relies on technical measures, leaving them vulnerable to systemic shocks, as evidenced by incidents like the 2016 DAO hack [8, 46]. In contrast, IT governance frameworks like COBIT and ISO/IEC 38500 incorporate systematic risk identification, monitoring, and mitigation through formal control mechanisms [34]. The most significant divergence lies in compliance. DAOs' pseudonymous and borderless nature makes them resistant to centralized oversight, often lacking Anti-Money Laundering (AML), Know Your Customer (KYC), and tax compliance mechanisms [4][21]. This poses particular challenges in countries like Indonesia, where regulators emphasize compliance frameworks grounded in centralized accountability [14][15]. IT governance frameworks, conversely, embed compliance as a core principle, creating a persistent gap that results in regulatory tensions in emerging markets.

B. Stakeholder Management Challenges in DAOs

Stakeholder management is a cornerstone of effective governance, ensuring that diverse interests are recognized, balanced, and incorporated into decision-making. In Decentralized Autonomous Organizations (DAOs), stakeholder management is particularly complex due to the absence of formal hierarchies and the reliance on token-weighted governance mechanisms. This challenge is amplified in Indonesia's DeFi ecosystem (see Table 2), where stakeholders include token holders, developers, exchanges, regulators, and retail investors. While DAOs provide new modes of coordination, their governance mechanisms expose significant limitations in aligning stakeholder interests with equitable participation and regulatory compliance.

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TABLE II. DAO GOVERNANCE CHALLENGES AND INDONESIAN CONTEXT

Challenge	Indonesian Implications
Power asymmetry	In Indonesia, most crypto participation is retail and speculative, making DAOs vulnerable to external dominance by global "whales." This may undermine local user influence and discourage grassroots governance.
Lack of participation	Indonesia's 17+ million crypto investors are largely retail traders, many with low financial literacy. This raises risks of governance apathy, scams, and susceptibility to misinformation in DAO participation.
Cross-jurisdictional complexity	Indonesia regulates crypto as a commodity under Bappebti and prohibits its use as payment. DAOs, however, operate globally without borders, making it unclear how taxation, AML/KYC, or liability rules apply to Indonesian participants.

DAO governance faces persistent challenges, particularly power asymmetry where token ownership dictates decision-making influence. This leads to power concentration among wealthy actors and institutional investors, as seen in Uniswap where venture capital firms hold significant sway, raising concerns about decentralization [7][32][47]. For Indonesian retail investors, who largely participate for speculative purposes with small token holdings [11], meaningful engagement in governance is severely limited. This dynamic exacerbates inequalities between global institutional players and local retail participants, undermining the democratic ideals of DAOs.

Another significant hurdle is low governance literacy, despite Indonesia's high crypto adoption rates [24]. While millennials and Gen Z are active in trading, their engagement in DAO governance discussions and proposals is minimal [20]. This mirrors a global trend of voter apathy in DAOs, driven by low incentives, technical barriers, or mistrust [18], [23]. In Indonesia, this issue is compounded by limited financial education and awareness of governance mechanics, making retail participants vulnerable to misinformation and governance capture, thus hindering their agency in DAO decision-making.

Furthermore, the integration of regulators as external stakeholders and the tension between global and local compliance present complex challenges. Indonesian regulators like Bappebti, OJK, and Bank Indonesia (BI) influence the DeFi ecosystem through policies, even without direct DAO participation [14]. However, DAOs are not designed to incorporate regulators, creating a disconnect between decentralized decision-making and national regulatory oversight. This governance gap is further complicated by the global nature of DAO governance clashing with territorial Indonesian regulations. Indonesian token holders might participate in votes that lead to non-compliant activities, exposing them to risks, such as engaging with decentralized lending protocols without OJK's consumer safeguards [16]. These conflicts highlight the structural misalignment between global DAO governance mechanisms and Indonesia's regulatory jurisdiction [8][21].

C. *Regulatory and Compliance Gaps in Indonesia*

The Indonesian regulatory landscape for digital assets remains fragmented and presents critical challenges when applied to decentralized autonomous organizations (DAOs). As can be seen from Table III, while regulators have established frameworks for centralized exchanges and crypto asset trading, DAOs fall largely outside of formal recognition, creating compliance ambiguities for participants and developers.

Indonesia's regulatory framework for crypto assets, while progressive in some areas, presents significant challenges for Decentralized Autonomous Organizations (DAOs). The Commodity Futures Trading Supervisory Agency (Bappebti) classifies crypto assets as commodities, regulating them similarly to futures contracts through licensed exchanges like Indodax and Tokocrypto [11]. However, this oversight does not extend to DAO-based protocols such as Aave or Compound, leaving them in a regulatory vacuum for Indonesian participants [4]. This absence of regulation creates a governance gap, exposing retail users to risks without institutional safeguards. Furthermore, Bank Indonesia (BI) prohibits the use of crypto assets as legal tender or payment instruments, reflecting a preference for monetary sovereignty and the development of a central bank digital currency (CBDC), the Digital Rupiah [48]. This directly conflicts with DAO-governed stablecoins like MakerDAO's DAI, which are designed for transactional use, illustrating a fundamental incompatibility between BI's centralized vision and DAO-driven monetary instruments [48].

The lack of clear regulatory authority also creates significant gaps in consumer protection and compliance. The Financial Services Authority (OJK), which supervises consumer protection in traditional financial sectors, has no authority over DAOs, leaving Indonesian users without formal recourse in cases of hacks, scams, or governance failures [20]. This exposes users to risks well-documented in global DeFi markets, including coding flaws, treasury mismanagement, and malicious actors leading to financial losses [21]. In parallel, Indonesia enforces strict anti-money laundering (AML) and know-your-customer (KYC) obligations, aligned with global standards. However, DAOs typically operate pseudonymous, allowing participation without identity verification, creating a direct compliance conflict [46]. While AI-driven monitoring tools are emerging in DeFi, most DAOs lack mechanisms to address AML/KYC, preventing regulators from enforcing financial integrity provisions and acting as a barrier to integrating DAOs into the formal financial system [46].

Finally, the existing tax framework struggles to accommodate the nuances of DAO activities. While Indonesia has imposed a 0.1% value-added tax (VAT) and a 0.1% income tax on crypto transactions since May 2022, signifying progress in formalizing the sector [16], DAO-based activities like yield farming, staking, or governance token rewards fall outside clear definitions. This ambiguity creates uncertainty for both regulators and users regarding tax liabilities. Global studies similarly highlight the difficulty of applying existing taxation frameworks to decentralized financial systems [2, 49]. Without explicit rules, Indonesian DAO participants face uncertainty, risking either under-compliance or double taxation in cross-border activities, further underscoring the need for a comprehensive regulatory approach to DAOs in the country.

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TABLE III. KEY INDONESIAN REGULATIONS TO DEFI AND DAOS

Regulators	Document / Year	Main Provision	Implication for DAOs
Bappebti (Commodity Futures Trading Supervisory Agency)	Peraturan Bappebti No. 5/2019 on Physical Market of Crypto Assets / 2019	Defines crypto assets as commodities; regulates trading via licensed futures exchanges and custodians.	DAO tokens may be treated as commodities, but decentralized protocols fall outside centralized exchange licensing.
Bank Indonesia (BI)	Peraturan BI No. 20/6/PBI/2018 on Financial Technology / 2018	Prohibits the use of crypto assets as legal payment instruments; emphasizes oversight of fintech innovations.	DAOs issuing or managing stablecoins (e.g., MakerDAO's DAI) may conflict with payment restrictions.
Bank Indonesia (BI)	White Paper: Digital Rupiah (Project Garuda)	Outlines the roadmap for Indonesia's Central Bank Digital Currency (CBDC).	Opens potential for regulated digital currency; DAOs could face competition or integration challenges.
Otoritas Jasa Keuangan (OJK)	Roadmap for Digital Financial Innovation 2023–2027 / 2023	Provides regulatory priorities for fintech, consumer protection, and systemic stability; introduces regulatory sandbox.	No explicit DAO regulation, but sandbox could enable controlled DAO experiments
Directorate General of Taxes, Ministry of Finance	Crypto Taxation Regulation (PMK No. 68/PMK.03/2022)	Imposes VAT (0.1%) and income tax (0.1%) on crypto transactions conducted through licensed exchanges.	DAO-based yield farming, staking, or governance rewards are not clearly addressed, leaving ambiguity for participants.

DAOs currently do not fit within Indonesia's legal framework. While regulatory bodies such as Bappebti, BI, and OJK provide oversight for centralized exchanges and fintech activities, decentralized governance structures remain unaddressed. Compliance requirements for AML, KYC, and taxation are unresolved, leaving DAO participants in legal ambiguity and exposing them to significant risks. This gap illustrates the broader regulatory paradox in Indonesia: the state seeks to harness digital finance innovation while maintaining centralized oversight, a tension that DAOs, by their very design, challenge [4][15].

D. Toward a Hybrid DAO–IT Governance Model for Indonesia

The findings from the mapping exercise indicate that while DAOs demonstrate strong transparency, they lack accountability, stakeholder alignment, risk management, and compliance when assessed against IT governance frameworks and Indonesia's regulatory requirements (Figure 2). To address these gaps, a hybrid governance model is necessary—one that preserves the decentralized ethos of DAOs while embedding principles from IT governance frameworks (COBIT, ITIL, ISO/IEC 38500) and aligning with Indonesian compliance structures.

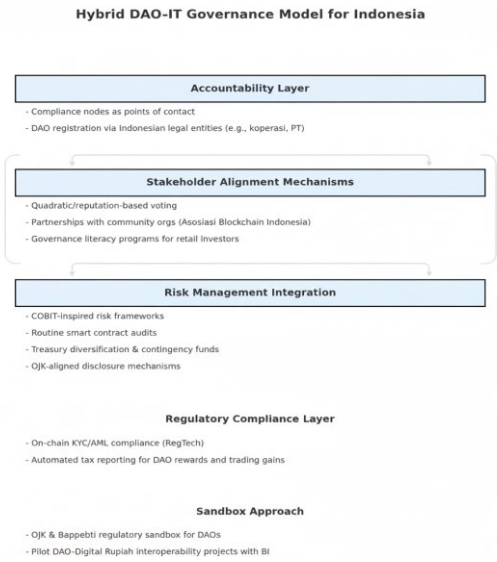

Fig. 2. Hybrid DAO–IT governance model for Indonesia.

The hybrid DAO–IT governance model addresses critical weaknesses in Decentralized Autonomous Organization (DAO) governance, particularly concerning accountability and regulatory compliance within the Indonesian context. To enhance accountability, the model proposes "compliance nodes" as formal regulatory touchpoints and suggests DAO registration under Indonesian legal entities like cooperatives or limited liability companies, aligning with ISO/IEC 38500 [14]. For regulatory compliance, it recommends incorporating RegTech solutions, such as on-chain KYC modules, to meet Indonesia's strict financial integrity requirements and aligning DAO activities with 2022 taxation frameworks through automated reporting [4][16]. This approach aims to provide clarity for both regulators and participants, reducing legal ambiguity and integrating DAOs into Indonesia's fiscal system.

Furthermore, the model tackles power asymmetry and underdeveloped risk management in DAOs. It suggests alternative voting mechanisms like quadratic or reputation-based governance to counter the dominance of large token holders and encourage broader retail investor participation, which is particularly relevant in Indonesia given the high rate of crypto adoption among retail investors [11][47][51]. Partnerships with community organizations like the Asosiasi Blockchain Indonesia are also proposed to improve governance literacy and mitigate voter apathy, aligning with COBIT's principle of stakeholder value. For risk management, the model advocates for adapting COBIT-inspired frameworks, including routine smart contract audits, treasury diversification, contingency funds, and disclosure mechanisms aligned with OJK's consumer protection guidelines, to inform Indonesian participants of potential risks [46].

Finally, the hybrid model emphasizes a "sandbox approach" under the supervision of OJK and Bappebti. This allows for controlled testing of DAO governance models, enabling regulators to observe risks and design appropriate

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policies [35]. Pilot projects could involve collaborations with BI's Digital Rupiah initiative to explore interoperability between decentralized assets and central bank digital currencies (CBDCs) [48]. This experimental approach aims to position Indonesia at the forefront of balancing decentralized innovation with centralized monetary stability, providing insights for other emerging markets grappling with similar challenges. By embedding accountability, enhancing stakeholder inclusivity, integrating risk management, and adopting RegTech-enabled compliance mechanisms, this model offers a pathway for DAOs to operate in alignment with both IT governance principles and Indonesia's legal requirements.

IV. Conclusion

This study investigated the governance challenges of Decentralized Finance (DeFi) through Decentralized Autonomous Organizations (DAOs), analyzing their alignment with IT governance principles and Indonesian regulatory compliance frameworks. Qualitative document analysis of academic literature, DAO governance documentation, IT governance standards, and Indonesian regulatory texts revealed several key insights. DAOs present a fundamentally different governance paradigm from traditional IT management, offering high transparency through on-chain records, which can surpass many centralized organizations. However, when measured against established IT governance frameworks like COBIT, ITIL, and ISO/IEC 38500, DAOs exhibit significant weaknesses in accountability, stakeholder alignment, risk management, and compliance, posing operational risks for DAO communities and systemic challenges when interacting with national regulators.

Stakeholder management within DAOs is problematic, particularly in Indonesia. While DAOs theoretically promote inclusive participation, token-weighted voting often leads to power imbalances, allowing large investors or "whales" to dominate decisions. Most Indonesian crypto users, who are retail investors, have limited holdings and low governance literacy, marginalizing them in DAO processes. Furthermore, key external stakeholders like Bappebti, Bank Indonesia, and OJK, whose policies are crucial for DAO viability, are not integrated into DAO designs, creating a disconnect between global decentralized communities and national regulatory authorities.

The Indonesian regulatory environment reveals a significant compliance gap. Bappebti's regulation of crypto as commodities primarily targets centralized exchanges, not DAOs. Bank Indonesia's prohibition of crypto as a payment instrument creates legal conflict for DAO-issued stablecoins. OJK's focus on consumer protection and systemic stability finds DAO participants lacking legal recourse under current Indonesian law. Additionally, while crypto transaction taxation has been in place since 2022, its application to DAO-based activities such as yield farming or governance rewards remains unclear, leaving Indonesian DAO participants in legal uncertainty. These findings collectively suggest that current DAOs cannot be smoothly integrated into Indonesia's financial and regulatory landscape, underscoring the need for a hybrid governance model that aligns DAO principles with IT governance best practices and Indonesian regulatory compliance.

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